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Statistical Deepfake Detection

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Abstract

The rapid progress of generative artificial intelligence has made it possible to create highly realistic synthetic images and videos. Technologies such as Generative Adversarial Networks (GANs), diffusion models, and transformer-based architectures can produce media that is often difficult to distinguish from authentic content. While these capabilities enable new creative and technical applications, they also introduce risks related to misinformation, identity impersonation, and digital fraud.

This thesis investigates statistical approaches for detecting AI-generated images using interpretable handcrafted features. Instead of relying solely on end-to-end deep learning models, the proposed method analyzes measurable statistical properties of images that differ between natural photographs and synthetic content. Features are extracted from several domains, including texture patterns, color distributions, frequency-domain representations, noise residuals, and intensity statistics.

The discriminative power of these features is evaluated using statistical analysis and Linear Discriminant Analysis (LDA). Experimental results show that certain feature families, particularly noise-based and frequency-based descriptors, provide strong signals for distinguishing real images from synthetic ones.

Artificial intelligence tools were used during the preparation of this thesis to assist with text organization, language refinement, and general structuring of written content. All technical analysis, experimental design, implementation, and results interpretation were performed by the author.

Keywords: Deepfake detection, image forensics, statistical features, frequency analysis, texture analysis, machine learning.

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Chapter 1

Introduction

1.1 Statistical Deepfake Detection

1.1.1 The Deepfake Challenge

Recent advances in generative artificial intelligence have made it possible to produce highly realistic synthetic media. Modern generative models such as Generative Adversarial Networks (GANs), diffusion models, and multi-modal transformer architectures can generate faces, voices, and videos that are nearly indistinguishable from authentic recordings.

While these technologies enable creative applications in entertainment, education, and digital content creation, they also introduce serious societal risks. Synthetic media can be used to spread misinformation, impersonate individuals, manipulate public opinion, or fabricate evidence. The ability to produce convincing deepfakes has therefore shifted the technological challenge from *generation* to *verification*: determining whether a piece of media is genuine or synthetically generated.

Because modern generation tools require little technical expertise and can produce realistic results within seconds, automated detection systems are essential for preserving trust in digital media. Deepfake detection therefore plays a critical role in maintaining the integrity of journalism, legal evidence, and online communication.

1.1.2 Global Awareness and Policy Response

Governments and international institutions have increasingly recognized deepfakes as a major technological and societal challenge. Regulatory initiatives in both the European Union and the United States have highlighted the need for reliable detection mechanisms.

The European Union has addressed synthetic media through legislation such as the **AI Act** and the **Digital Services Act**, which emphasize transparency and accountability in AI-generated content. In addition, the **Code of Practice on Disinformation** identifies deepfakes as a key threat to democratic processes.

Similarly, in the United States, organizations such as the **Department of Homeland Security** and the **Federal Trade Commission** have launched initiatives aimed at combating synthetic media misuse. Legislative proposals such as the **DEEPFAKES Accountability Act** also emphasize the need for mechanisms that can verify the authenticity of digital media.

These developments demonstrate that deepfake detection is no longer only a technical research problem but also a critical component of digital governance and cybersecurity.

1.1.3 Detection Approaches

Most modern deepfake detection methods rely on deep neural networks trained as binary classifiers. These models, typically based on convolutional neural networks or transformers, learn to distinguish real and synthetic images directly from raw pixel data.

Although such approaches can achieve high accuracy on benchmark datasets, they often suffer from several limitations. In particular, detectors trained on one dataset frequently fail when applied to media generated by previously unseen models or when images are heavily compressed. This phenomenon, known as *dataset bias*, occurs because neural networks may learn superficial visual patterns specific to a dataset rather than fundamental properties of real images.

In addition, purely deep learning approaches often operate as black boxes,

making it difficult to interpret their decisions. This lack of interpretability can be problematic in forensic contexts, where explanations for detection decisions may be required.

1.1.4 Role of Statistical Methods

Statistical approaches provide an alternative perspective on deepfake detection. Instead of focusing on high-level visual semantics, these methods analyze the underlying statistical properties of images.

Natural images follow characteristic statistical regularities that arise from physical processes such as camera sensors, optical systems, and scene illumination. Synthetic images produced by generative models often fail to reproduce these properties perfectly. As a result, deepfakes frequently exhibit subtle inconsistencies that can be detected through statistical analysis.

Several categories of statistical cues have been explored in the literature:

- **Frequency-domain artifacts**, which arise from non-physical upsampling and generative priors used in neural image synthesis.
- **Sensor-noise inconsistencies**, particularly deviations in Photo-Response Non-Uniformity (PRNU), which acts as a fingerprint of real camera sensors.
- **Noise-residual statistics**, capturing irregularities in the fine-grained noise patterns present in authentic images.
- **Natural image distribution laws**, such as the statistical behavior of gradients, textures, and transform coefficients.

These statistical cues are typically low-dimensional and interpretable, allowing them to be analyzed using classical statistical techniques or lightweight machine learning models.

1.1.5 Advantages of Statistical Detection

Statistical approaches offer several advantages that complement modern deep learning detectors.

First, they are often **model-agnostic**. Because they rely on fundamental properties of natural images rather than dataset-specific patterns, statistical cues can generalize better across different generative architectures.

Second, statistical detectors tend to be **computationally efficient**. Many methods operate on compact feature vectors derived from image transforms or residual maps, making them suitable for real-time screening or deployment on limited hardware.

Third, statistical cues provide a degree of **interpretability**. When a detection decision is based on measurable properties such as abnormal frequency distributions or inconsistent noise patterns, it becomes easier to explain why a given image is likely to be synthetic.

For these reasons, statistical analysis plays an important role in digital media forensics and is often used alongside deep learning detectors in practical systems.

1.1.6 Hybrid Detection Architectures

Recent research increasingly combines statistical analysis with deep neural networks. These hybrid architectures integrate explicit statistical features into deep models in order to improve robustness and generalization.

Examples include frequency-aware convolutional networks that process both spatial and spectral representations of images, as well as detectors that explicitly analyze noise residuals before performing classification. By constraining neural networks to focus on meaningful statistical artifacts, such hybrid approaches aim to reduce overfitting and improve reliability across different datasets.

1.1.7 Overview of Statistical Features

In this work, statistical analysis of images plays a central role in the detection pipeline. Rather than relying solely on end-to-end deep models, the proposed approach focuses on extracting interpretable image descriptors that capture structural and statistical properties of visual data.

These descriptors include features derived from image kernels, frequency transforms, and global distribution statistics. By transforming complex image structures into numerical features, statistical techniques can be applied to evaluate which characteristics differ significantly between real and synthetic images.

The following sections introduce the statistical tools used in this thesis, including feature extraction techniques and the statistical tests employed to analyze their discriminative power. While detailed mathematical formulations will be presented later, the goal at this stage is to establish the intuition that deepfake detection can be approached as a statistical problem: identifying measurable deviations between the distributions of authentic and synthetic media.

Chapter 2

Evolution of Deepfake Generation Architectures

2.1 Introduction

Deepfake technology relies on generative models capable of synthesizing highly realistic images and videos. Over the past decade, several deep learning architectures have contributed to the development of deepfake systems. Each new paradigm introduced improvements in image realism, training stability, or modeling capacity.

The development of deepfake generation methods followed a clear chronological progression. Early systems relied on autoencoders to perform identity transfer between faces. Generative Adversarial Networks (GANs) later enabled high-resolution and photorealistic image synthesis. Diffusion models introduced a fundamentally different generative paradigm based on iterative denoising, leading to significant improvements in visual quality and diversity. More recently, transformer architectures have been incorporated to model temporal consistency and multimodal interactions in video deepfakes.

This chapter presents these architectures in chronological order, explaining both their mathematical foundations and their role in the evolution of deepfake technology.

2.2 Generative Modeling Framework

Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the image space and let $\mathbb{P}_r$ represent the real data distribution. A generative model aims to learn a distribution $\mathbb{P}_g$ such that

$$\mathbb{P}_g \approx \mathbb{P}_r$$

Once trained, the model generates synthetic samples

$$x \sim \mathbb{P}_g$$

that resemble real images. Different architectures approximate $\mathbb{P}_r$ using different mathematical strategies, including reconstruction objectives, adversarial optimization, stochastic processes, and attention mechanisms.

2.3 Autoencoders: The First Deepfake Systems

2.3.1 Historical Context

The first publicly available deepfake tools in 2017 relied on autoencoder architectures. These systems enabled face swapping by learning a shared representation of facial expressions and pose while maintaining identity-specific decoding networks.

Although autoencoders had been studied earlier in representation learning, their application to face-swapping marked the beginning of practical deepfake technology.

2.3.2 Mathematical Formulation

An autoencoder consists of two functions:

$$E_\phi : \mathcal{X} \rightarrow \mathcal{Z}$$

$$D_\theta : \mathcal{Z} \rightarrow \mathcal{X}$$

where $\mathcal{Z}$ is a latent space representing compressed features of the input image.

The training objective minimizes the reconstruction error

$$\mathcal{L}_{rec} = \mathbb{E}_{x \sim \mathbb{P}_r} [\|x - D_\theta(E_\phi(x))\|^2]$$

2.3.3 Face Swapping Architecture

For identities A and B , early deepfake systems used a shared encoder E with identity-specific decoders D_A and D_B :

$$\mathcal{L} = \mathbb{E}_{x_A \sim \mathbb{P}_A} [\|x_A - D_A(E(x_A))\|^2] + \mathbb{E}_{x_B \sim \mathbb{P}_B} [\|x_B - D_B(E(x_B))\|^2]$$

Face swapping occurs by decoding a latent representation using the opposite decoder

$$x_{A \rightarrow B} = D_B(E(x_A))$$

2.3.4 Limitations

Although autoencoders enabled the first deepfakes, they suffered from several limitations:

- low image resolution
- blurry outputs
- limited realism
- poor generalization

These limitations motivated the transition toward adversarial learning.

2.4 Generative Adversarial Networks

2.4.1 Adversarial Learning

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. in 2014, significantly improved image realism. GANs frame generative modeling as a game between two networks:

- generator G
- discriminator D

The generator produces synthetic images

$$G : \mathcal{Z} \rightarrow \mathcal{X}$$

while the discriminator attempts to distinguish real images from generated ones.

The objective function is

$$\min_G \max_D V(G, D) = \mathbb{E}_{x \sim \mathbb{P}_r} [\log D(x)] + \mathbb{E}_{z \sim \mathbb{P}_z} [\log(1 - D(G(z)))]$$

2.4.2 Distribution Matching

For a fixed generator, the optimal discriminator is

$$D_G^*(x) = \frac{p_r(x)}{p_r(x) + p_g(x)}$$

Training a GAN therefore minimizes the Jensen–Shannon divergence between $\mathbb{P}_r$ and $\mathbb{P}_g$.

2.4.3 Wasserstein GAN

Training instability led to the development of Wasserstein GANs, which minimize the Wasserstein distance

$$W_1(\mathbb{P}_r, \mathbb{P}_g) = \sup_{\|f\|_L \leq 1} (\mathbb{E}_{x \sim \mathbb{P}_r}[f(x)] - \mathbb{E}_{z \sim \mathbb{P}_g}[f(G(z))])$$

This formulation provides smoother gradients and improved training stability.

2.4.4 Impact on Deepfakes

GAN architectures dramatically improved deepfake realism. Models such as StyleGAN enabled the generation of high-resolution, photorealistic faces.

GANs became widely used for

- face synthesis
- identity transfer
- expression reenactment
- image enhancement

However, GAN training remains unstable and can suffer from mode collapse. These challenges encouraged the exploration of alternative generative frameworks.

2.5 Diffusion Models

2.5.1 Generative Process

Diffusion models represent a fundamentally different approach to generative modeling. Instead of learning to generate images directly, the model learns to reverse a stochastic noise process.

The forward diffusion process gradually adds Gaussian noise:

$$dx = f(x, t)dt + g(t)dw$$

where w is a Wiener process.

2.5.2 Reverse Diffusion

Generation corresponds to reversing the diffusion process

$$dx = [f(x, t) - g(t)^2 \nabla_x \log p_t(x)] dt + g(t) d\bar{w}$$

The term $\nabla_x \log p_t(x)$ is the score function that indicates the direction toward higher data density.

2.5.3 Score Matching Objective

A neural network $s_\theta(x, t)$ approximates the score function using the loss

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} [\|s_\theta(x_t, t) - \nabla_{x_t} \log p_{t|0}(x_t|x_0)\|^2]$$

2.5.4 Advantages

Diffusion models offer several advantages:

- improved training stability
- high visual fidelity
- strong diversity of generated samples

2.5.5 Modern Applications

Diffusion models power many modern generative systems such as Stable Diffusion and DALL-E. In the context of deepfakes, they enable

- text-driven face generation
- image editing and inpainting
- identity-controlled synthesis

2.6 Transformers in Deepfake Generation

2.6.1 Self-Attention

Transformers rely on the self-attention mechanism to model relationships between elements of an input sequence.

For input embeddings

$$X \in \mathbb{R}^{n \times d}$$

the attention mechanism is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

2.6.2 Video Modeling

Transformers are particularly effective for modeling temporal relationships in video sequences.

For frame t , the representation can be computed as

$$x_t = \text{Attention}(q_t, K_{context}, V_{context})$$

This mechanism allows information to be shared across multiple frames.

2.6.3 Multimodal Deepfake Systems

Modern deepfake systems often combine visual and audio inputs. Transformers enable this multimodal integration through cross-attention mechanisms.

For example, lip synchronization can be modeled as

$$P(y_{mouth}|x_{audio}, x_{face}) = \text{Decoder}(\text{CrossAttention}(x_{audio}, x_{face}))$$

2.6.4 Current Role

Today, transformers are frequently integrated with diffusion or GAN-based architectures to provide

- temporal consistency in videos
- long-range dependency modeling
- audio-visual synchronization

2.7 Evolution of Deepfake Architectures

Architecture	Period	Key Innovation	Usage Today
Autoencoders	2017	Shared latent representation for identity transfer	Mostly historical; used in early face-swapping tools
GANs	2017–2020	Adversarial training for photorealistic synthesis	Still used for face synthesis and refinement
Diffusion Models	2020–present	Iterative denoising generation	Dominant paradigm for high-quality image generation
Transformers	2021–present	Self-attention and multimodal modeling	Used for video coherence and multimodal generation

Table 2.1: Evolution of deepfake generation architectures

2.8 Summary

The development of deepfake technology reflects the broader evolution of generative modeling in deep learning. Early systems relied on autoencoders for identity transfer but produced limited image quality. GANs introduced adversarial learning and significantly improved realism. Diffusion models further improved generation quality and stability through stochastic denoising

processes. Finally, transformer architectures enabled modeling of temporal dependencies and multimodal interactions.

Modern deepfake systems often combine multiple architectures, such as diffusion models enhanced with transformer-based attention mechanisms, to achieve highly realistic and temporally consistent video synthesis.

Chapter 3

Statistical Features

3.1 Introduction

Detecting AI-generated images requires identifying statistical differences between natural photographs and synthetically generated content. While deep learning approaches can automatically learn representations, handcrafted features remain valuable because they provide interpretable statistical descriptors of image structure.

In this work, handcrafted features are used to characterize multiple aspects of image statistics, including spatial texture patterns, color distributions, frequency-domain representations, and basic intensity properties. Each image is therefore represented by a vector of numerical descriptors capturing complementary visual characteristics.

The purpose of this chapter is to present the mathematical foundations and implementation procedures of the feature families used in this thesis.

3.2 Feature Representation

Let an image be represented as a function

$$I : \Omega \rightarrow \mathbb{R}^3$$

where $\Omega \subset \mathbb{R}^2$ represents the spatial image domain and the output cor-

responds to RGB color intensities.

After applying feature extraction procedures, each image is mapped to a feature vector

$$\mathbf{x} \in \mathbb{R}^d$$

where d represents the dimensionality of the extracted feature space. Each dimension corresponds to a specific statistical descriptor derived from the image.

These features aim to capture differences in statistical structure between real and AI-generated images.

3.3 Texture Features

Texture features describe spatial patterns and local relationships between pixels. Natural images typically contain complex textures formed by physical structures, while AI-generated images may produce synthetic textures with different statistical properties.

Three complementary texture descriptors are used: Local Binary Patterns, Gray-Level Co-occurrence Matrices, and Gabor filters.

3.3.1 Local Binary Patterns

Local Binary Patterns capture local intensity relationships around a central pixel:

$$\text{LBP}_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c) 2^p$$

$$s(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

3.3.2 Texture Feature Extraction Algorithm

Algorithm 1 Texture Feature Extraction

Require: Input image I

Ensure: Texture feature vector $x_{texture}$

- 1: Convert I to grayscale
 - 2: Compute Local Binary Pattern map
 - 3: Compute histogram of LBP codes
 - 4: Compute Gray-Level Co-occurrence Matrix
 - 5: Extract statistical descriptors
 - 6: Apply Gabor filter bank
 - 7: Compute mean and variance of filter responses
 - 8: Concatenate all descriptors into $x_{texture}$
 - 9: **return** $x_{texture}$
-

Feature Stage Showcase (Side-by-Side REAL vs FAKE): TEXTURE



Figure 3.1: Visualization of extracted texture features.

3.4 Color Features

Color features describe the statistical distribution of color intensities across different color spaces.

$$\mu_c = \frac{1}{N} \sum_{i=1}^N I_c(i)$$

$$\sigma_c = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (I_c(i) - \mu_c)^2}$$

Cross-channel correlations are also considered:

$$\rho_{c_1, c_2} = \frac{\sum_i (I_{c_1}(i) - \mu_{c_1})(I_{c_2}(i) - \mu_{c_2})}{\sqrt{\sum_i (I_{c_1}(i) - \mu_{c_1})^2 \sum_i (I_{c_2}(i) - \mu_{c_2})^2}}$$

3.4.1 Color Feature Extraction

Algorithm 2 Color Feature Extraction

Require: Image I

Ensure: Color feature vector x_{color}

- 1: Convert image to RGB, HSV and LAB
 - 2: **for** each channel **do**
 - 3: Compute mean
 - 4: Compute standard deviation
 - 5: **end for**
 - 6: Compute cross-channel correlations
 - 7: Concatenate descriptors
 - 8: **return** x_{color}
-

Feature Stage Showcase (Side-by-Side REAL vs FAKE): COLOR

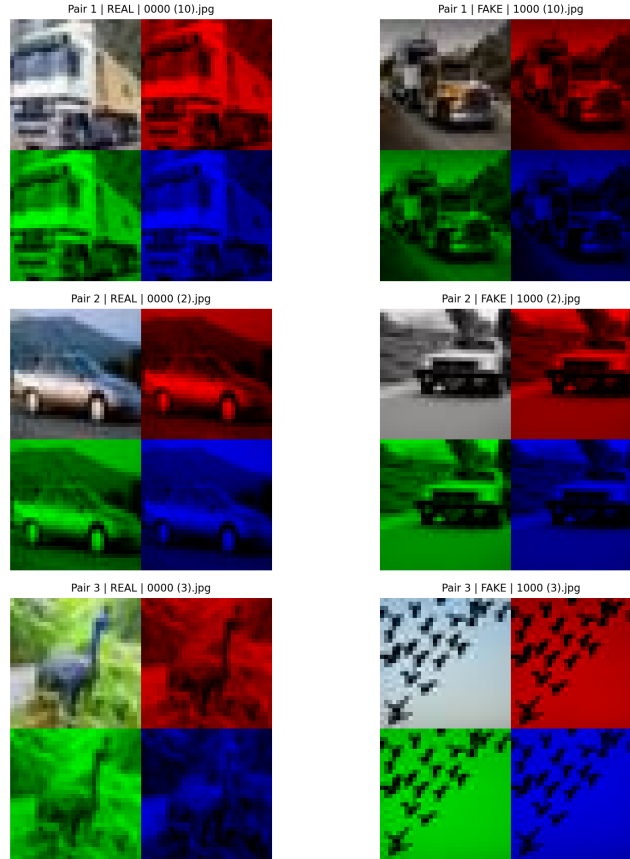


Figure 3.2: Extracted color statistics across color spaces.

3.5 Frequency Domain Features

3.5.1 Fourier Transform Features

The two-dimensional Fourier Transform converts spatial image data into frequency components.

3.5.2 FFT Feature Extraction

Algorithm 3 FFT Feature Extraction

Require: Image I

Ensure: Frequency feature vector x_{fft}

- 1: Convert to grayscale
 - 2: Compute 2D FFT
 - 3: Shift spectrum to center
 - 4: Compute magnitude spectrum
 - 5: Extract statistical descriptors
 - 6: **return** x_{fft}
-

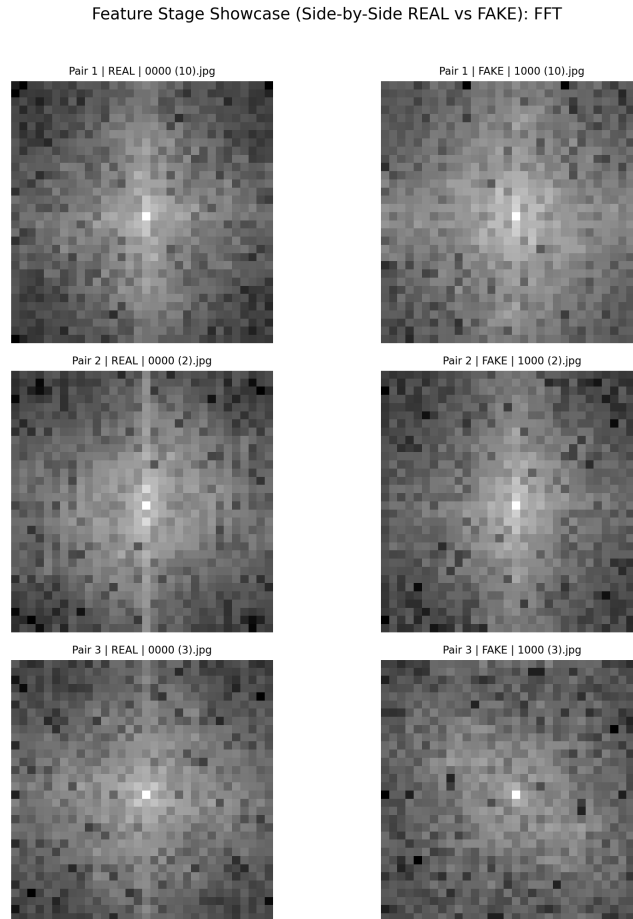


Figure 3.3: Frequency spectrum obtained from FFT.

3.5.3 Discrete Cosine Transform

$$F(u, v) = \alpha(u)\alpha(v) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cos \left[\frac{(2x+1)u\pi}{2M} \right] \cos \left[\frac{(2y+1)v\pi}{2N} \right]$$

3.5.4 DCT Feature Extraction

Algorithm 4 DCT Feature Extraction

Require: Image I

Ensure: Feature vector x_{dct}

- 1: Convert image to grayscale
 - 2: Compute 2D DCT
 - 3: Separate low and high frequencies
 - 4: Compute mean, variance, skewness, kurtosis
 - 5: Concatenate descriptors
 - 6: **return** x_{dct}
-

Feature Stage Showcase (Side-by-Side REAL vs FAKE): DCT

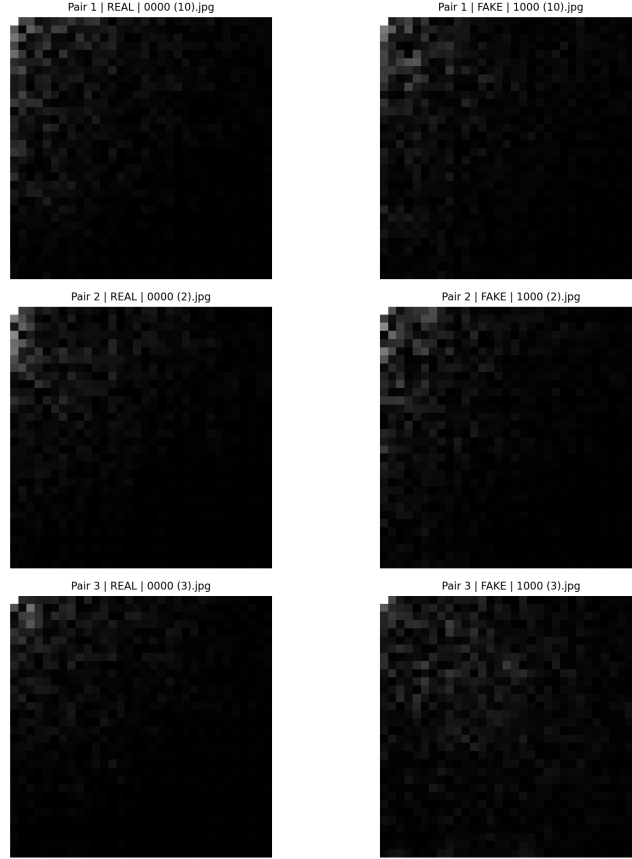


Figure 3.4: DCT frequency representation of the image.

3.5.5 Wavelet Features

Wavelet transforms provide a multi-resolution representation of images.

$$cA = (f * \phi) \downarrow 2$$

$$cH = (f * \psi_h) \downarrow 2$$

$$cV = (f * \psi_v) \downarrow 2$$

$$cD = (f * \psi_d) \downarrow 2$$

3.5.6 Wavelet Feature Extraction

Algorithm 5 Wavelet Feature Extraction

Require: Image I

Ensure: Wavelet feature vector $x_{wavelet}$

- 1: Convert image to grayscale
 - 2: Apply discrete wavelet transform
 - 3: Extract subbands cA, cH, cV, cD
 - 4: Compute energy statistics
 - 5: Concatenate descriptors
 - 6: **return** $x_{wavelet}$
-

Feature Stage Showcase (Side-by-Side REAL vs FAKE): WAVELET

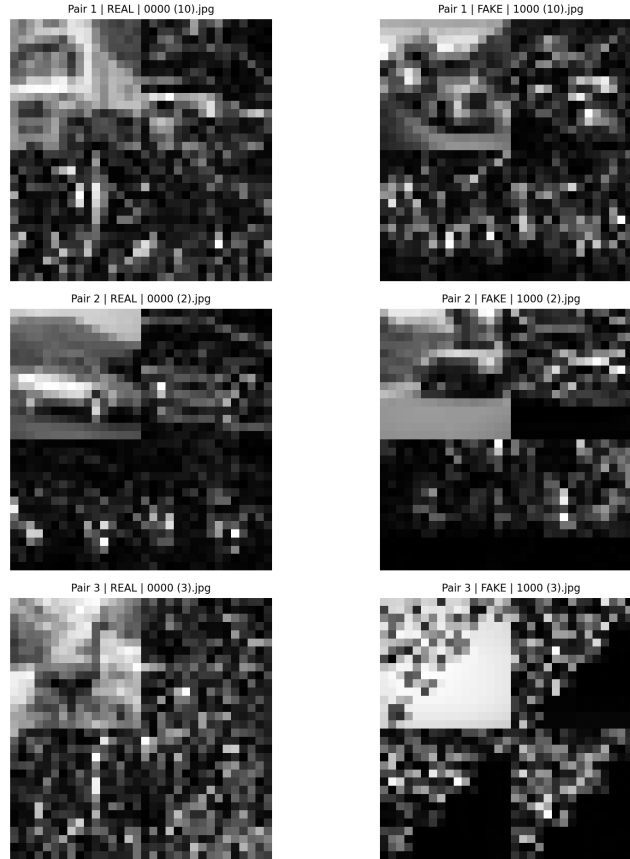


Figure 3.5: Wavelet decomposition showing multi-scale frequency structure.

3.6 Noise Features

Noise residuals capture inconsistencies in the fine-grained noise structure of images.

3.6.1 Noise Feature Extraction

Algorithm 6 Noise Residual Extraction

Require: Image I

Ensure: Noise feature vector x_{noise}

- 1: Apply denoising filter
 - 2: Compute residual $R = I - \hat{I}$
 - 3: Compute statistical descriptors of R
 - 4: Concatenate descriptors
 - 5: **return** x_{noise}
-

Feature Stage Showcase (Side-by-Side REAL vs FAKE): NOISE

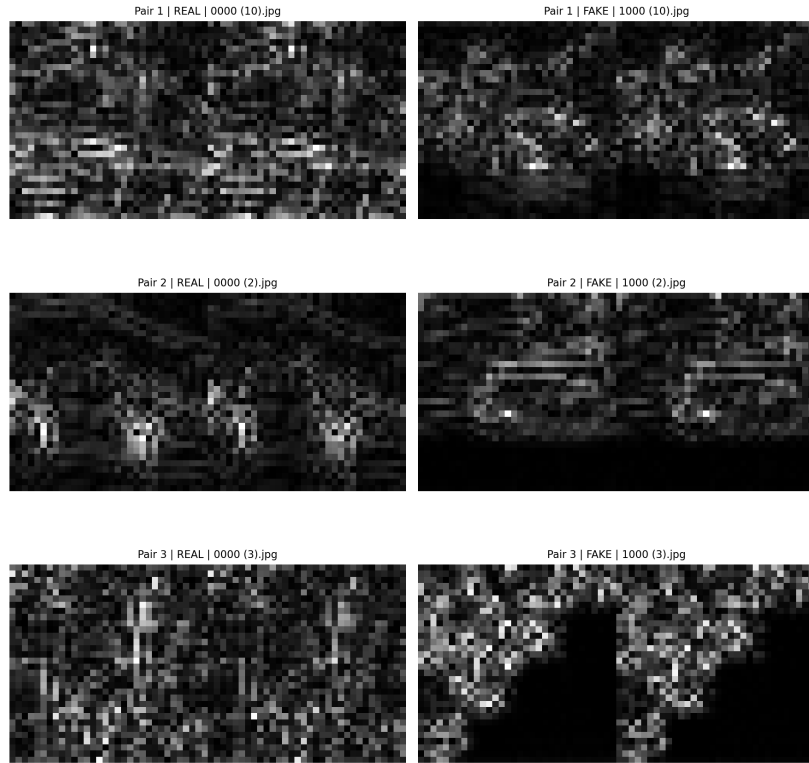


Figure 3.6: Noise residual visualization extracted from the image.

3.7 Intensity Features

First-order statistical descriptors are extracted from grayscale intensities.

$$\mu = \frac{1}{N} \sum_i g_i$$

$$\sigma^2 = \frac{1}{N-1} \sum_i (g_i - \mu)^2$$

$$\gamma_1 = \frac{\frac{1}{N} \sum_i (g_i - \mu)^3}{\sigma^3}$$

$$\gamma_2 = \frac{\frac{1}{N} \sum_i (g_i - \mu)^4}{\sigma^4} - 3$$

$$H = - \sum_{k=0}^{255} p_k \log_2 p_k$$

3.7.1 Intensity Feature Extraction

Algorithm 7 Intensity Feature Extraction

Require: Image I

Ensure: Intensity feature vector $x_{intensity}$

- 1: Convert image to grayscale
 - 2: Compute mean, variance
 - 3: Compute skewness and kurtosis
 - 4: Compute entropy
 - 5: Concatenate descriptors
 - 6: **return** $x_{intensity}$
-

Feature Stage Showcase (Side-by-Side REAL vs FAKE): INTENSITY

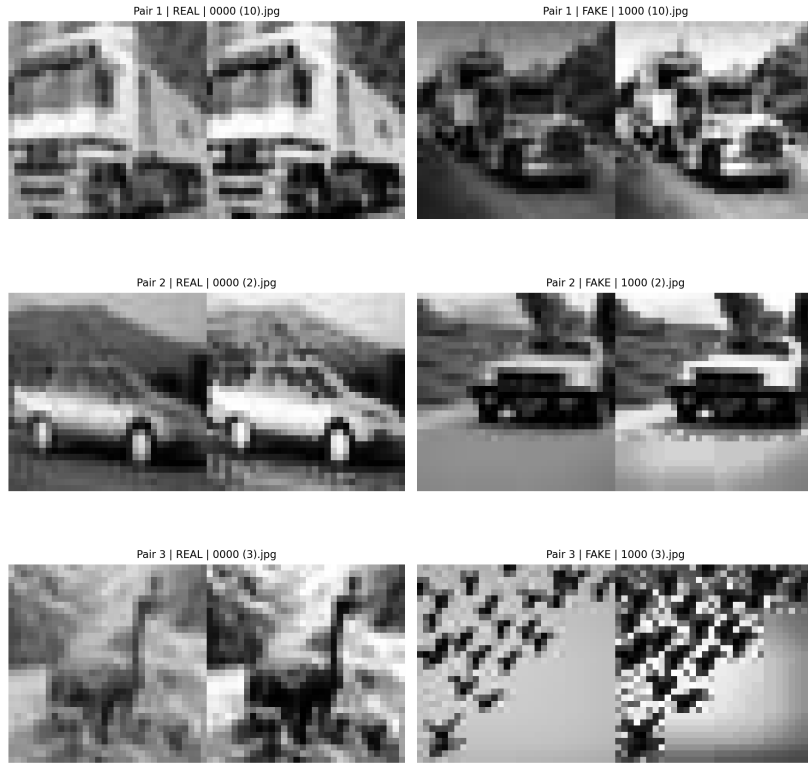


Figure 3.7: Visualization of intensity-based statistics.

3.8 Feature Summary

Table 3.1: Feature Categories and Dimensionality

Feature Type	Dimensions
Texture	64
Color	42
DCT	32
Wavelet	28
Intensity	24

Each feature family captures a different aspect of image statistics, providing a diverse representation of visual characteristics.

3.9 Conclusion

This chapter introduced the handcrafted feature extraction techniques used in this thesis. By combining texture descriptors, color statistics, frequency-domain representations, and intensity-based measures, a comprehensive feature representation of images is obtained.

These features serve as the foundation for the statistical analysis and classification methods presented in the following chapters.

3.10 Experimental Results

The effectiveness of the proposed statistical feature representation was evaluated on the **CIFAKE dataset**, a large-scale dataset containing both real and AI-generated images. The experiments were conducted on approximately **60,000 images**, including both authentic photographs and synthetic samples generated by modern generative models.

For each image, the feature extraction pipeline described in the previous sections was applied in order to compute descriptors from multiple domains, including texture statistics, color distributions, frequency-domain representations, wavelet coefficients, noise residuals, and intensity statistics.

To evaluate the discriminative power of each feature family, two complementary metrics were used:

- **Covariance Difference** – measures how strongly the statistical distributions of real and synthetic images differ in the feature space.
- **Linear Discriminant Analysis (LDA) Accuracy** – evaluates how well the features support linear classification between the two classes.

The covariance difference was computed using the Frobenius norm of the difference between the covariance matrices of the real and synthetic feature

distributions. This metric captures structural differences in feature variability and correlations.

LDA accuracy measures the classification performance obtained by projecting the feature vectors onto a direction that maximizes the separation between the two classes while minimizing intra-class variance.

Table 3.2: Feature Evaluation on the CIFAKE Dataset

Feature Type	Cov Diff (Frobenius)	LDA Accuracy
Noise	2.0487×10^3	0.8482
FFT	6.2333×10^{11}	0.7987
Texture	1.0470×10^5	0.7624
Color	5.2340×10^3	0.7299
DCT	4.6505×10^3	0.6823
Intensity	2.6082×10^3	0.6432
Wavelet	8.9893×10^3	0.6308

The results reveal several important observations.

First, **noise-based features achieved the highest classification performance**, reaching an LDA accuracy of 0.8482. This indicates that synthetic images exhibit noticeable inconsistencies in their noise patterns when compared to real photographs, which naturally contain sensor noise produced by camera hardware.

Second, **FFT features produced the largest covariance difference**, indicating strong statistical divergence between the spectral properties of real and synthetic images. However, their slightly lower LDA accuracy suggests that these differences are not purely linearly separable and may be better exploited by non-linear classifiers.

Texture and color features also provide meaningful discriminative signals, capturing differences in local spatial structure and color relationships introduced by generative models.

Overall, the results confirm that **statistical image features contain significant information for distinguishing real images from AI-generated content**. In particular, noise-based and frequency-domain descriptors appear to be the most informative components of the feature representation.

These findings demonstrate that interpretable statistical descriptors can provide a strong foundation for deepfake detection, complementing modern deep learning approaches.

Chapter 4

Experimental Evaluation

4.1 Dataset

The evaluation of the proposed feature extraction framework was conducted using the **CIFAKE dataset**. CIFAKE is a large-scale dataset designed for deepfake detection research and contains both real images and synthetic images generated using modern generative models.

For the experiments in this thesis, approximately **60,000 images** from the dataset were analyzed. The dataset contains a balanced mixture of authentic photographs and AI-generated images, allowing a direct comparison of their statistical properties.

Each image was processed using the feature extraction pipeline described in the previous chapter. This pipeline extracts statistical descriptors from multiple domains of image representation, including texture patterns, color distributions, frequency-domain features, wavelet decompositions, noise residuals, and intensity statistics.

4.2 Evaluation Metrics

To evaluate the effectiveness of each feature family in distinguishing real and synthetic images, two complementary metrics were used.

4.2.1 Covariance Difference

The covariance difference measures how strongly the statistical distributions of real and synthetic images differ in the feature space.

Let Σ_{real} and Σ_{fake} denote the covariance matrices computed from the feature vectors of real and synthetic images. The difference between these distributions is quantified using the Frobenius norm:

$$||\Sigma_{real} - \Sigma_{fake}||_F$$

A larger covariance difference indicates stronger structural differences between the feature distributions of the two classes. This metric captures differences in feature variability and correlations and can reveal statistical discrepancies even when the classes are not linearly separable.

4.2.2 Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) was used to evaluate the classification performance of each feature family.

LDA finds a projection of the feature space that maximizes the separation between classes while minimizing intra-class variance. The resulting classification accuracy provides an estimate of how well the features support linear discrimination between real and synthetic images.

Higher LDA accuracy indicates stronger discriminative power.

4.3 Results

The results of the evaluation are summarized in Table 4.1. Each feature family was analyzed independently in order to measure its contribution to the detection task.

Table 4.1: Feature Evaluation on the CIFAKE Dataset

Feature Type	Cov Diff (Frobenius)	LDA Accuracy
Noise	2.0487×10^3	0.8482
FFT	6.2333×10^{11}	0.7987
Texture	1.0470×10^5	0.7624
Color	5.2340×10^3	0.7299
DCT	4.6505×10^3	0.6823
Intensity	2.6082×10^3	0.6432
Wavelet	8.9893×10^3	0.6308

The results highlight several important observations regarding the statistical behavior of real and AI-generated images.

First, **noise-based features achieved the highest classification performance**, reaching an LDA accuracy of 0.8482. This indicates that synthetic images exhibit noticeable inconsistencies in their noise patterns when compared to real photographs. Real images naturally contain sensor noise produced by camera hardware, while synthetic images often fail to reproduce these noise characteristics accurately.

Second, **FFT features produced the largest covariance difference**. This suggests that the spectral distributions of real and synthetic images differ substantially in the frequency domain. However, the slightly lower LDA accuracy indicates that these differences are not perfectly linearly separable. This observation suggests that FFT features may be better exploited by non-linear classifiers.

Texture features also demonstrated strong performance, achieving an LDA accuracy of 0.7624. These features capture local spatial patterns and micro-texture characteristics that may differ between real images and generative outputs.

Color features provided moderate discriminative power, indicating that generative models can often approximate global color statistics reasonably well, although subtle differences remain detectable.

Finally, wavelet and intensity features exhibited lower classification performance compared to the other feature families, although they still captured

measurable statistical differences between the two classes.

4.4 Discussion

The experimental results confirm that statistical image features contain significant information for distinguishing real images from AI-generated content.

In particular, **noise and frequency-domain features appear to be the most informative descriptors**. Noise residuals capture inconsistencies in sensor noise patterns, while frequency-based representations reveal spectral artifacts introduced by generative models.

These findings suggest that combining multiple feature families, especially noise, FFT, and texture descriptors, may provide a robust feature representation for deepfake detection systems.

Overall, the results demonstrate that interpretable statistical descriptors can serve as a strong foundation for detecting synthetic media and can complement modern deep learning approaches in practical detection pipelines.

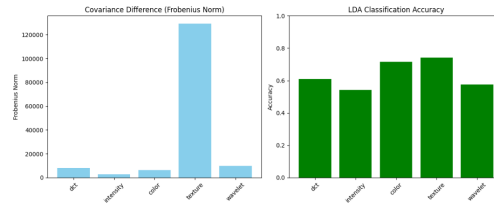


Figure 4.1: Comparison of feature families based on covariance difference and LDA accuracy.

Bibliography